Finite Element Formulation for Linear Sloshing in a Rectangular Tank

1 Introduction

This document outlines the theoretical foundations for the finite element method (FEM) applied to linear sloshing of an incompressible, inviscid fluid inside a rigid container. The formulation is based on potential flow theory and its linearization for small-amplitude free-surface oscillations.

2 Governing Equations

For an incompressible and inviscid fluid, the velocity field is expressed through a scalar potential:

$$\boldsymbol{v} = \nabla\phi, \tag{1}$$

where $\phi(x, y, t)$ is the velocity potential. Incompressibility requires

$$\nabla^2\phi = 0 \quad \text{in the fluid domain.} \tag{2}$$

The pressure field follows from the linearized Bernoulli equation:

$$p' = -\rho\frac{\partial\phi}{\partial t} - \rho g\,\eta, \tag{3}$$

where $\eta(x, t)$ is the vertical displacement of the free surface.

3 Boundary Conditions

3.1 Rigid Walls

On the walls and bottom, the fluid cannot cross the boundary:

$$\frac{\partial\phi}{\partial n} = 0. \tag{4}$$

3.2 Linearized Free-Surface Boundary Conditions

Two boundary conditions apply on the free surface ($y = H$):

1. Kinematic condition:
$$\frac{\partial\eta}{\partial t} = \frac{\partial\phi}{\partial y}. \tag{5}$$
2. Dynamic condition:
$$\frac{\partial\phi}{\partial t} + g\,\eta = 0. \tag{6}$$

Combining the two yields a single evolution equation for η:

$$\frac{\partial^2\eta}{\partial t^2} + g\,\frac{\partial\phi}{\partial y} = 0. \tag{7}$$

4 Reduction to a Free-Surface FEM Problem

For small-amplitude sloshing, it is common to eliminate ϕ and derive a boundary integral or a reduced free-surface FEM equation. A classical approach leads to the eigenvalue problem:

$$\boldsymbol{K}\boldsymbol{\eta} = \omega^2 \boldsymbol{M}\boldsymbol{\eta}, \tag{8}$$

where $\boldsymbol{\eta}$ is the nodal vector of free-surface elevations.

The matrix $\boldsymbol{K}$ comes from gravitational restoring effects:

$$K_{ij} = \rho g \int_{A_e} \nabla N_i \cdot \nabla N_j \; dA, \tag{9}$$

with N_i the shape functions of triangular CST elements.

The mass matrix, assuming lumping, is

$$M_{ii} = \rho \frac{A_e}{3}, \tag{10}$$

for each node of a triangular element of area A_e.

5 Element Matrices

For a triangular element with nodes (i, j, k), area

$$A = \frac{1}{2} |x_i(y_j - y_k) + x_j(y_k - y_i) + x_k(y_i - y_j)|, \tag{11}$$

the gradients of the shape functions are expressed using standard CST coefficients:

$$b_1 = y_j - y_k, \qquad c_1 = x_k - x_j, \tag{12}$$

$$b_2 = y_k - y_i, \qquad c_2 = x_i - x_k, \tag{13}$$

$$b_3 = y_i - y_j, \qquad c_3 = x_j - x_i. \tag{14}$$

The stiffness matrix becomes:

$$K_{ab} = \rho g \, t \frac{b_a b_b + c_a c_b}{4A}, \tag{15}$$

where t is the out-of-plane thickness.

The lumped mass matrix is:

$$M = \rho t A \begin{bmatrix} 1/3 & 0 & 0 \\ 0 & 1/3 & 0 \\ 0 & 0 & 1/3 \end{bmatrix}. \tag{16}$$

6 Eigenvalue Problem and Modal Analysis

The generalized eigenproblem

$$\boldsymbol{K}\boldsymbol{\eta} = \omega^2 \boldsymbol{M}\boldsymbol{\eta} \tag{17}$$

leads to natural frequencies

$$f = \frac{\omega}{2\pi} \tag{18}$$

and periods

$$T = \frac{1}{f}. \tag{19}$$

The eigenvectors describe the shape of the free-surface oscillations. The pressure fluctuations inside the fluid follow:

$$p'(x, y) = \rho g \, \eta(x). \tag{20}$$

7 References

- Faltinsen, O. M., *Sea Loads on Ships and Offshore Structures.* Cambridge Univ. Press, 1990.
- Mei, C. C., *The Applied Dynamics of Ocean Surface Waves.* World Scientific, 1989.
- Bathe, K.-J., *Finite Element Procedures.* Prentice Hall, 1996.
- Zienkiewicz, O. C., Taylor, R. L., *The Finite Element Method.* McGraw-Hill.

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8 Governing Equations (Expanded)

For an incompressible, inviscid fluid, the flow is irrotational and can be described by a velocity potential $\phi(x, y, t)$:

$$\boldsymbol{v} = \nabla\phi, \tag{21}$$

where $\boldsymbol{v}$ is the fluid velocity vector. The incompressibility condition leads to Laplace's equation in the fluid domain Ω:

$$\nabla^2\phi = \frac{\partial^2\phi}{\partial x^2} + \frac{\partial^2\phi}{\partial y^2} = 0, \quad \text{in } \Omega. \tag{22}$$

On the rigid boundaries (walls and bottom), the normal component of velocity must vanish:

$$\frac{\partial\phi}{\partial n} = 0. \tag{23}$$

At the free surface $y = H$, the boundary conditions are linearized for small-amplitude oscillations:

1. Kinematic condition:
$$\frac{\partial\eta}{\partial t} = \frac{\partial\phi}{\partial y}\Big|_{y=H}, \tag{24}$$
where $\eta(x, t)$ is the vertical displacement of the free surface.

2. Dynamic condition:
$$\frac{\partial\phi}{\partial t} + g\,\eta = 0, \quad y = H. \tag{25}$$

By differentiating the dynamic condition with respect to time and substituting the kinematic condition, one obtains a single second-order differential equation for the free-surface displacement $\eta(x, t)$:

$$\frac{\partial^2\eta}{\partial t^2} + g\,\frac{\partial\phi}{\partial y}\Big|_{y=H} = 0. \tag{26}$$

In a variational (FEM) formulation, this is typically converted into a weak form over the fluid domain, leading to a generalized eigenvalue problem for the nodal free-surface elevations:

$$\boldsymbol{K\eta} = \omega^2\boldsymbol{M\eta}, \tag{27}$$

where $\boldsymbol{\eta}$ is the vector of nodal elevations (spostamento verticale del pelo libero), $\boldsymbol{M}$ is the mass matrix, and $\boldsymbol{K}$ is the stiffness matrix derived from the gravitational restoring force.

9 Connection Between PDE and FEM Matrices

The stiffness matrix $\boldsymbol{K}$ represents the discrete version of the linear operator associated with the free-surface restoring force:

$$K_{ij} = \rho g \int_{A_e} \nabla N_i \cdot \nabla N_j \; dA, \tag{28}$$

where N_i and N_j are the element shape functions, and A_e is the element area. Essentially, $\nabla N_i \cdot \nabla N_j$ arises from the weak form of the Laplace equation projected onto the free-surface displacement field η.

The mass matrix $\boldsymbol{M}$ represents the inertial term $\partial^2 \eta / \partial t^2$ in the PDE, typically assembled in lumped or consistent form:

$$M_{ii} = \rho t A_e / 3 \quad \text{for each node of a triangular element.} \tag{29}$$

Thus, the FEM eigenvalue problem directly approximates the PDE for the free-surface displacement. The eigenvectors provide the mode shapes of the surface, and the eigenvalues determine the natural sloshing frequencies.